

Sarvesh Patham

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CAREER OBJECTIVE

Improve industry experience in robotics, VLSI + FPGA applications, and embedded systems through a **summer internship**

EDUCATION

University of Illinois Urbana Champaign

Urbana, IL

Bachelor of Sciences in Electrical Engineering, Minor in Computer Science

Expected: May 2028

- Cumulative GPA: 3.9

EXPERIENCE

ARM

Summer 2026

Physical Implementation Intern

Austin, TX

- Developed backtracing CLI tool for detecting integrated clock gates in any technology mapped netlist
- Created large RTL hierarchy processing flow to translate hierarchies into easily traversable directed acyclic graphs
- Verified block-level physical designs and debugged bottlenecks utilizing Cadence Innovus and Synopsys Verdi

Electrical Engineer Intern

June 2025 – Present

General Autonomy AI

Piscataway, NJ

- Developed LiDAR/Camera/IMU data collection stack using NVIDIA Jetson Orin for simulator development
- Implemented autonomy stack using ROS 2 capable of lanelet planning and object detection from sensor data
- Secured \$50k in investments through LiDAR and Camera mount optimization for investor demos.

Undergraduate Research Assistant

August 2025 – May 2026

University of Illinois Urbana Champaign

Urbana, IL

- Constructed magnetic field verification system using Python, successfully field tested simulation field lines
- Designed hall effect sensor mount and transceiver + pre-amp housing using Fusion360
- Researching transceiver designs for decreasing signal-to-noise ratio and increasing data clarity for low field MRI

Mechanical Engineering Intern

October 2023 – April 2024

3D Pets

Bloomington, NJ

- Machined/constructed prosthetics to specific consumer standards; increased production rate by 10 percent
- Manufactured body molds and developed 3D Models for prosthetics by scanning molds with LiDAR scanner
- Developed an organizational standard for parts for a newly constructed workshop to efficiently increase workflow

PROJECTS/EXTRACURRICULARS

SigRobotics | *Electrical Engineer*

February 2026 – Present

- Determined stepper motor specifications, sensors, and gantry design for an industrial use autosampler
- Controlling autonomous sampling system via PLC in C++ for remote access through Ignition
- Designing pinch valves in Fusion360 for tubing based on industry specifications for analog controlled dispense

Shazam Song Matcher | *FPGA/Embedded Project*

March 2026 – May 2026

- Developed a Shazam-style matcher that captured live audio and generated frequency hashes in SystemVerilog
- Used MicroBlaze C in Vitis to read BRAM hashes, compare against a database, and output matches over UART
- Built Python tools for UART audio streaming, song recording, and fingerprint database generation

Cuff-Link | *Embedded Systems Project*

December 2023 – May 2024

- Designed a muscle and motion tracking device using Fusion360 and developed in the Arduino IDE
- Programmed python based virtual cursor controlled by onboard IMU and Electromyography sensor (EMG)
- Selected as a top 3 class finalist for school competition; demonstrated product to audience of 400+

TECHNICAL SKILLS

Digital: Python, Robotics Operating System(ROS 2), MATLAB, Linux, C,C++, HTML /CSS, CAD (Fusion360/OnShape), Arduino, ESP32

Hardware:PCB Design(KiCAD,Altium), LTspice/SPICE, SystemVerilog, Verilog, VHDL, Oscilloscopes, Function Generators, Soldering, Communication Protocols (UART/I2C/CAN)